

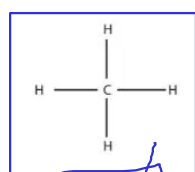
Química Orgânica

Nomenclatura dos hidrocarbonetos

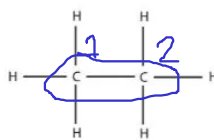
- Prefixo:** Indica o número de carbonos presentes na cadeia principal;
- Infixo:** Indica o tipo de ligação encontrada na cadeia;
- Sufixo:** Indica a função orgânica dos hidrocarbonetos terminando com a letra "o".

Prefixo	Infixo	Sufixo
1 C - MET	AN - Ligação Simples	
2 C - ET	EN - Ligação dupla	
3 C - PROP	IN - Ligação tripla	
4 C - BUT		
5 C - PENT		
6 C - HEX	DIEN - Duas ligações duplas	
7 C - HEPT		
8 C - OCT	DIIN - Duas ligações triplas	
9 C - NON		
10 C - DEC		

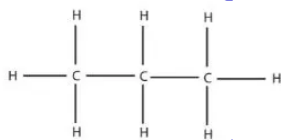
Alcanos



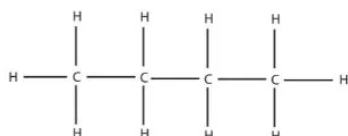
Metano



Etano

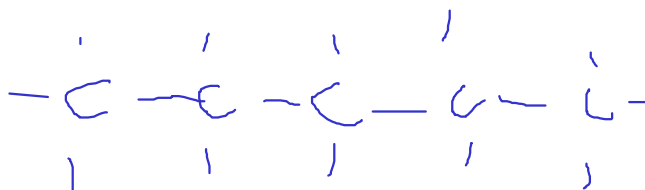


Propano

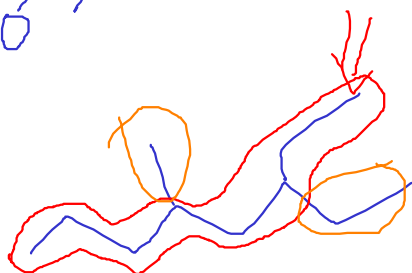


Butano

PENTANO

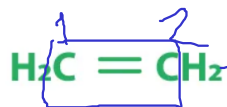


METANO



Alcénos

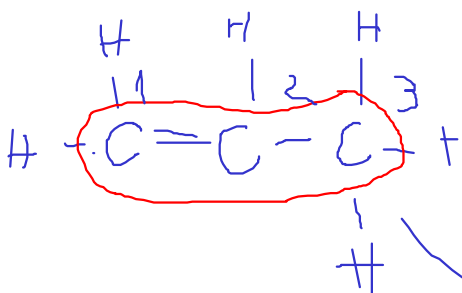
ETENO



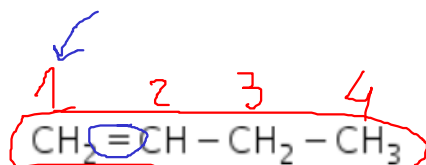
eteno
(etileno)



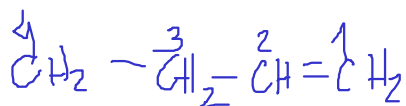
propeno
(propileno)



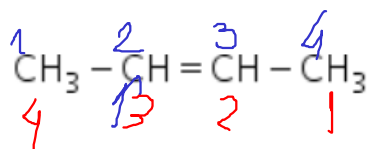
PROPENO



BUT-1-ENO

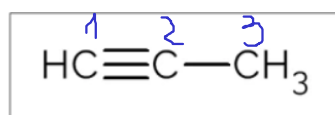


BUT-3-ENO

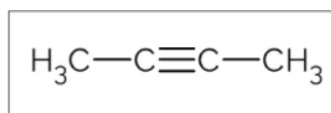


BUT-2-ENO

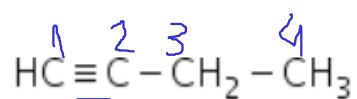
Alcinos → IN



propino



but-2-ino



BUT-1-INO

Hidrocarbonetos ramificados

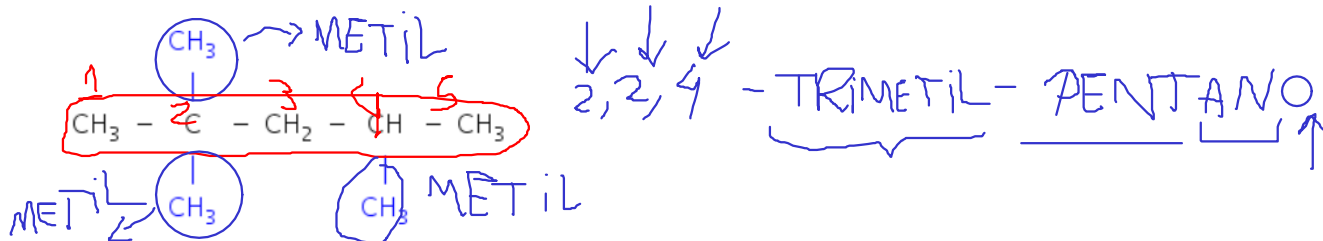
1. Quando existem duas ou mais ramificações iguais elas são indicadas usando os prefixos di, tri, tetra, etc.

-il

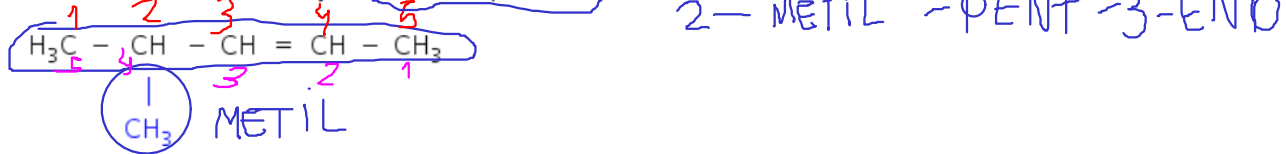
2. Quando as ramificações são diferentes elas são citadas em ordem alfabética.

3. No caso de alcenos e alcinos, a cadeia principal é aquela que apresenta a dupla ou tripla ligação. A numeração da cadeia deve se iniciar da extremidade mais próxima da ligação, independente das ramificações.

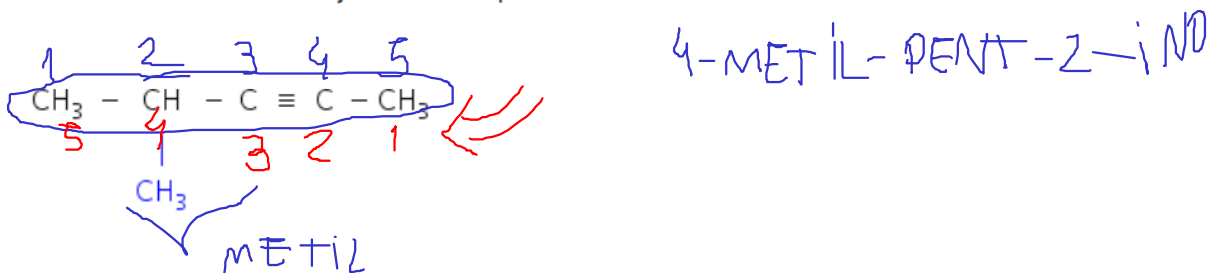
1. Alcano com ramificação: 2.2.4-trimetil-pentano



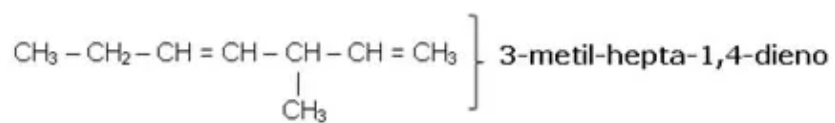
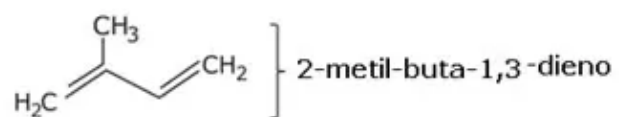
2. Alcano com ramificação: 4-metil-2-penteno



3. Alcano com ramificação: 4-metil-pent-2-ino



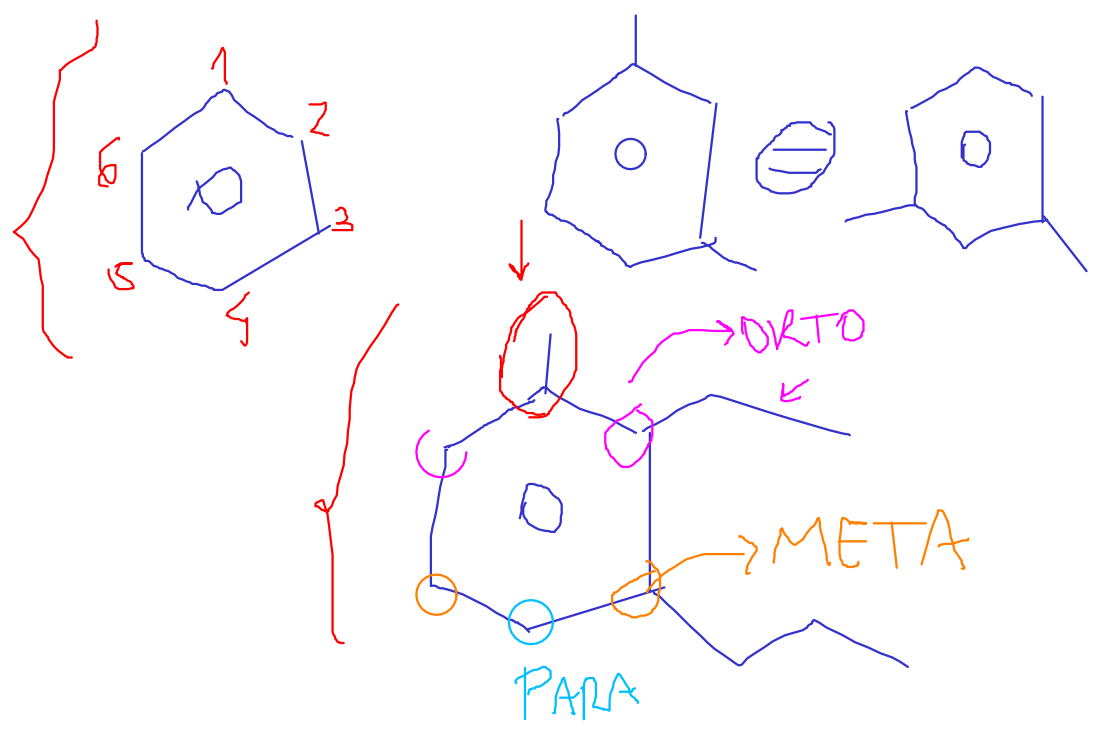
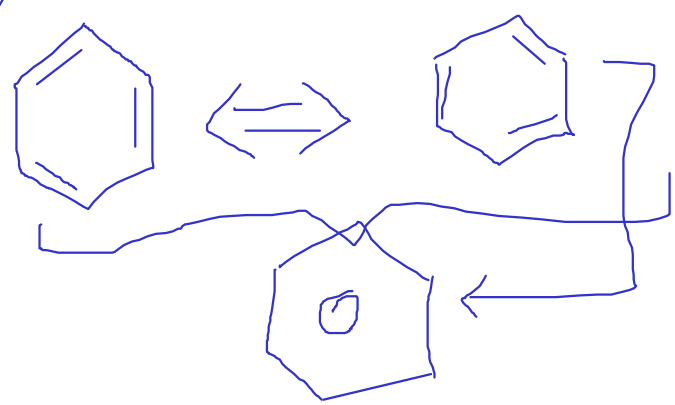
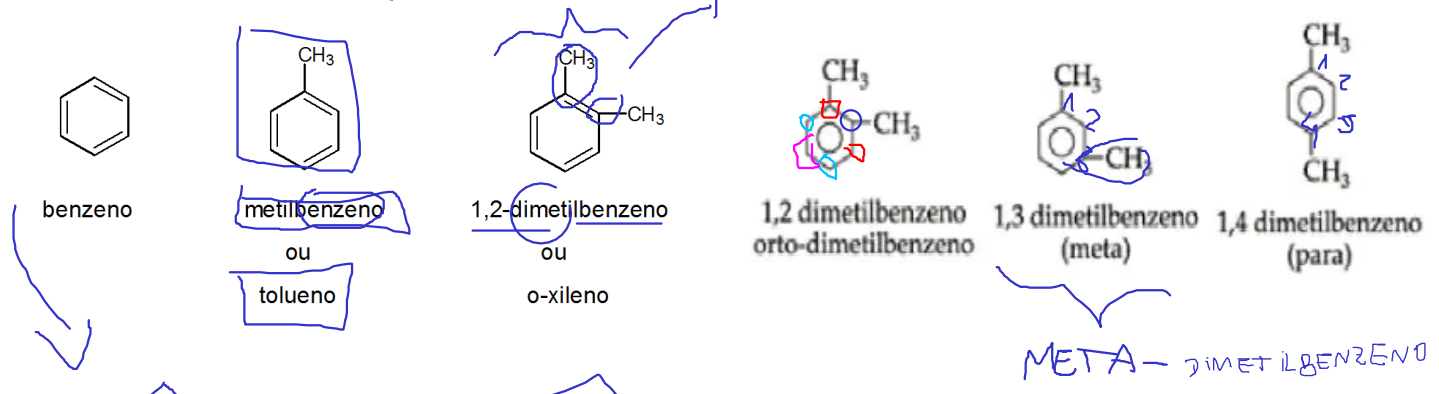
Alcadienos



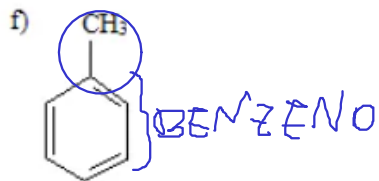
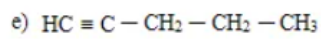
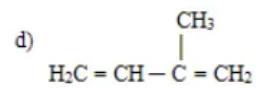
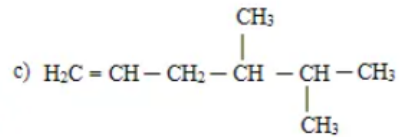
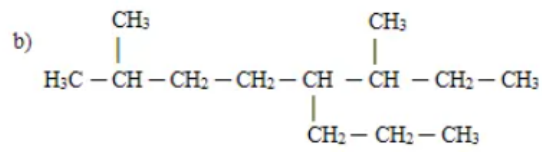
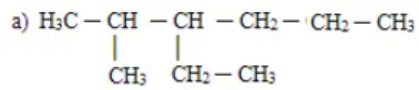
Buta-1,3-dieno ou 1,3-butadieno

Hidrocarbonetos aromáticos

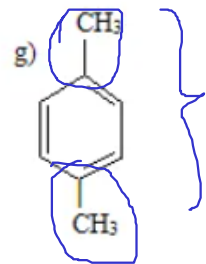
ressonância
ORTO-DIMETILBENZENO



Exercícios



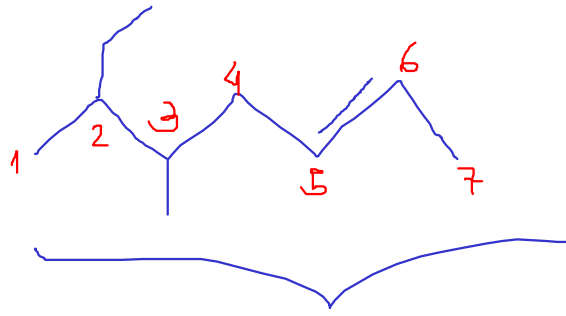
METIL BENZENO / TOLUENO



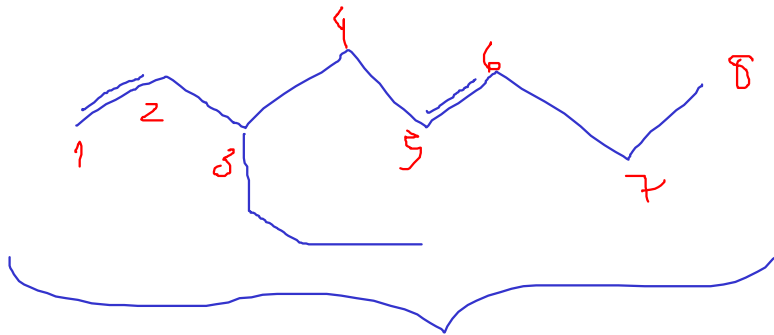
1,4 - DIMETILBENZENO

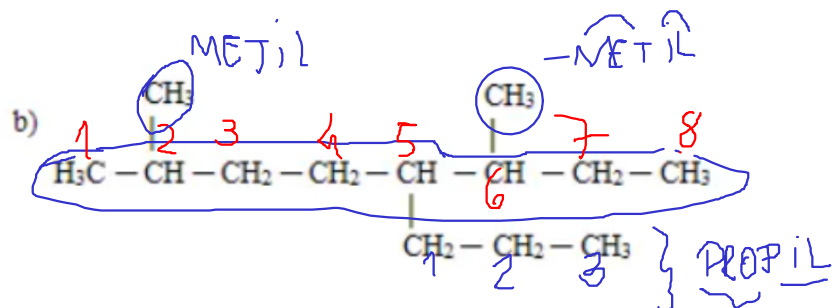
PARADIMETIL BENZENO

2-ETIL-3-METIL-HEPT-5-ENO

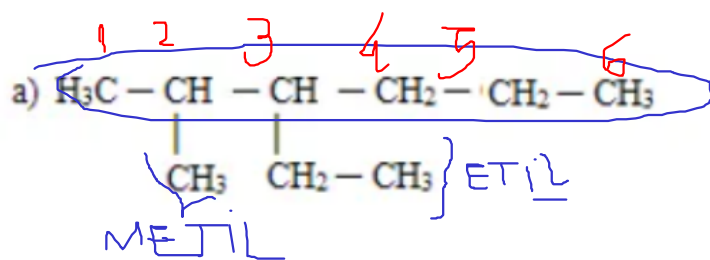


3-PROPIL-OCT-1,5-DIENO

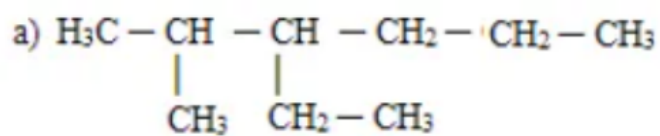


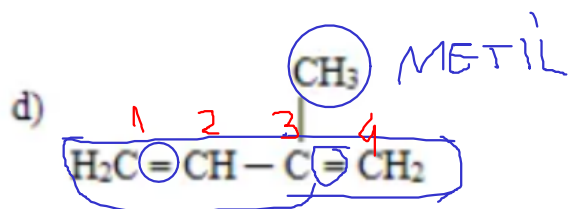


2,6-DIMETIL-5-PROPIL - OCTANO



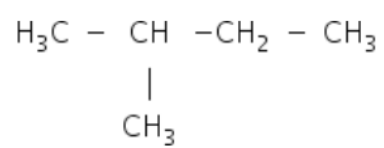
3-ETIL-2-METIL-HEXANO





3-METIL - BUT - 1,3 - DIENO

O nome correto do hidrocarboneto ramificado, cuja fórmula está esquematizada a seguir é:



Construa a estrutura do hidrocarboneto com as indicações abaixo.

- A cadeia principal contém 5 átomos de carbono;
- Todas as ligações na cadeia principal são simples;
- Existem 3 radicais metil na estrutura: dois no carbono 2 e um no carbono 4.

A nomenclatura oficial do composto é:

RAMIF

2,2,4 -TRIMETIL-PENTANO

a) trimetilpentano

b) 2,2,4 metilpentano

c) 2,2 metilpent-4-ano

d) 2,2,4 trimetilpentano

